

Revisiting Hierarchical Processing for CF mMIMO: A Novel Antenna-Aware Weighting Approach

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Abstract—This paper studies uplink cell-free massive multiple-input multiple-output networks with hierarchical processing, where distributed multi-antenna access points (APs) generate local soft estimates and the central processing unit (CPU) weighs these estimates to perform the final decoding. The existing large-scale fading decoding (LSFD) weighting scheme relies on scalar per-AP-user weights, which is inherently coarse for multi-antenna APs. We propose an antenna-aware weighting strategy that assigns a matrix-valued long-term weight to each AP-user pair, thereby generalizing the LSFD from scalar coefficients to antenna-aware matrix-valued weighting. The optimal bilinear equalization (OBE) solution is derived, which is shown to simultaneously maximize the spectral efficiency (SE) and minimize the conditional decoding mean-square error. Numerical results demonstrate that the proposed OBE weighting outperforms the LSFD in terms of SE, with SE gains that increase with the number of antennas and are particularly pronounced when the local maximum-ratio combining with limited interference-suppression capability is utilized.

I. INTRODUCTION

Multiple-input multiple-output (MIMO) technology is a key component of the latest wireless communication systems, offering substantial gains in spectral efficiency (SE) and link reliability by leveraging spatial multiplexing and beamforming [1]–[3]. In particular, massive MIMO has achieved remarkable success by exploiting large antenna arrays to approach favorable propagation and channel hardening, making it a cornerstone of fifth-generation (5G) networks. Alongside the continually increasing demand for higher data rates and ubiquitous coverage for wireless networks, the next stage of MIMO evolution has attracted significant attention [4]–[6]. Beyond the conventional evolution idea of scaling the antenna count, recent research increasingly emphasizes the flexibility and adaptability of MIMO technology, where network topologies and antenna resources are reconfigured to better match propagation conditions and traffic dynamics [7]–[11].

Within this broader trend, cell-free massive MIMO (CF mMIMO) stands out as a flexible network deployment architecture in which many geographically distributed access points (APs) coherently serve the user equipments (UEs) under coordination by the central processing unit (CPU) [12]–[15]. CF mMIMO leverages macro-diversity to deliver a more

uniform service quality across the entire coverage area. A key feature of CF mMIMO is its hierarchical processing structure, where each AP performs soft decoding in a local manner, and the CPU completes the final decoding by fusing soft information from the APs using long-term channel statistics. The second-layer fusion at the CPU can be implemented in different ways. A widely adopted and powerful approach is the large-scale fading decoding (LSFD), where the CPU assigns a dedicated scalar weight to each AP-UE pair and combines the locally decoded soft estimates accordingly. These LSFD coefficients can be optimized using channel statistics to effectively manage inter-user interference and to maximize the use-and-then-forget (UatF) bound-based achievable SE [12]. However, the LSFD can substantially outperform the average weighting with only a modest additional processing overhead. As a result, the LSFD strategy has received considerable attention and has become a widely adopted weighting strategy in the CF mMIMO literature [16]–[18]. Although LSFD is appealing from a complexity and scalability perspective, its weighting mechanism remains inherently coarse-grained because it assigns only a scalar weight to represent the overall contribution of each AP for a given UE. When the APs employ multi-antenna arrays, such a scalar coefficient cannot differentiate the roles of individual antenna elements, and thus the weighting operation cannot fully exploit the multi-antenna processing capability available within each AP. This structural restriction becomes more pronounced as the array size grows. Motivated by above observations, in this paper, we develop an antenna-aware weighting strategy that generalizes LSFD from scalar coefficients to matrix-valued weights. The proposed design follows the optimal bilinear equalization (OBE) philosophy [16] and preserves the hierarchical processing paradigm, while containing LSFD as a special case when the matrix weight reduces to a scaled identity.

The main contributions of this paper are as follows. First, we propose an antenna-aware OBE weighting strategy for uplink CF mMIMO networks by assigning a matrix-valued long-term weight to each AP-UE pair, which generalizes the conventional LSFD weighting from scalar coefficients to antenna-element-aware weighting while remaining compatible with arbitrary local combining vectors. Second, we derive an optimal OBE solution and show that it not only maximizes the UatF-based achievable SE, but also minimizes the conditional decoding mean square error (MSE). Numerical results under

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correlated Rician channels demonstrate that the OBE weighting consistently outperforms the LSFD weighting, with gains increasing with the number of antennas.

II. SYSTEM MODEL

In this paper, we consider an uplink CF mMIMO network in which M APs equipped with uniform planar arrays (UPAs) and connected to a CPU through fronthaul links. Each UPA contains $N = N_H N_V$ antennas, where N_H and N_V denote the numbers of antennas along the horizontal and vertical directions, respectively. The APs jointly serve K single-antenna UEs located within a given coverage region. A three-dimensional Cartesian coordinate system is adopted to describe the geometry. The coverage area is assumed to lie in the $x - z$ plane, and the $x - y$ plane is perpendicular to the coverage area, where the x -axis is parallel to the coverage area while the y -axis is perpendicular to it. Accordingly, the y -axis coordinate for the coverage area is 0. The bottom-left corner of the m -th array is located at $\mathbf{r}_m = [x_m, y_m, z_m]^T \in \mathbb{R}^3$ with $y_m = L_{AP}$ representing the array height. By indexing antenna elements row-by-row starting from the bottom-left corner, the coordinate of the n -th antenna element at the m -th array is given by $\mathbf{r}_{mn} = [r_{mn,x}, r_{mn,y}, r_{mn,z}]^T = [x_m + \text{mod}(n-1, N_H)\Delta_H, L_{BS} + [(n-1)/N_H]\Delta_H, z_m]^T \in \mathbb{R}^3$, where Δ_H and Δ_V denote the horizontal and vertical antenna spacing, respectively. Moreover, the coordinate of UE k is denoted by $\mathbf{s}_k = [s_{k,x}, s_{k,y}, s_{k,z}]^T \in \mathbb{R}^3$, with $s_{k,y} = L_{UE}$ being the UE height. We define $\mathbf{g}_{mk} \in \mathbb{C}^N$ as the channel response between the m -th AP array and UE k . A standard block-fading model is adopted, where $\mathbf{g}_{mk}$ is constant over each coherence block of length τ_c and varies independently across different coherence blocks [19]. Moreover, channels corresponding to different AP-UE pairs are assumed mutually independent. We focus on the correlated Rician fading channel model where

$$\mathbf{g}_{mk} = \bar{\mathbf{g}}_{mk} e^{j\theta_{mk}} + \check{\mathbf{g}}_{mk} \quad (1)$$

with $\bar{\mathbf{g}}_{mk} \in \mathbb{C}^N$ and $\check{\mathbf{g}}_{mk} \in \mathbb{C}^N$ being the deterministic line-of-sight (LoS) and random non-line-of-sight (NLoS) components, respectively, and $\theta_{mk} \sim \mathcal{U}(-\pi, \pi]$ representing the random phase shift on the LoS path. The NLoS component is modeled as $\check{\mathbf{g}}_{mk} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{R}_{mk})$ with $\mathbf{R}_{mk} \in \mathbb{C}^{N \times N}$ denoting the spatial correlation matrix between AP m and UE k .

In the channel estimation phase, τ_p mutually orthogonal pilot sequences are transmitted, where τ_p also represents the number of symbols allocated for pilot transmission within each coherence block. Let $\mathcal{P}_k$ denote the set of UEs that reuse the same pilot as UE k , including UE k itself. By applying the standard phase-aware minimum mean-square error (MMSE) estimation fundamentals in [12], the phase-aware MMSE estimate of $\mathbf{g}_{mk} \in \mathbb{C}^N$ at array m can be written as $\hat{\mathbf{g}}_{mk} = \bar{\mathbf{g}}_{mk} e^{j\theta_{mk}} + \sqrt{p_k} \mathbf{R}_{mk} \Psi_{mk}^{-1} (\mathbf{y}_{mk}^p - \bar{\mathbf{y}}_{mk}^p)$, where $\mathbf{y}_{mk}^p = \sum_{l \in \mathcal{P}_k} \sqrt{p_l} \tau_p \mathbf{g}_{ml} + \mathbf{n}_{mk}^p$, $\bar{\mathbf{y}}_{mk}^p = \sum_{l \in \mathcal{P}_k} \sqrt{p_l} \tau_p \bar{\mathbf{g}}_{ml} e^{j\theta_{ml}}$, $\mathbf{n}_{mk}^p \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \tau_p \sigma^2 \mathbf{I}_N)$ is the projected noise with σ^2 being the noise power, $\Psi_{mk} = \sum_{l \in \mathcal{P}_k} p_l \tau_p \mathbf{R}_{ml} + \sigma^2 \mathbf{I}_N$, and p_k denotes the uplink transmit power of UE k . Additional details on phase-aware MMSE estimation are provided in [16].

III. UPLINK TRANSMISSION AND OBE WEIGHTING

We focus on uplink data transmission and introduce a new OBE-based weighting strategy to improve the achievable SE compared with the conventional LSFD weighting approach in [12]. Each UE transmits $\tau_u = \tau_c - \tau_p$ data symbols in each τ_c -length coherence block used for uplink transmission. The received signal at AP m is

$$\mathbf{y}_m = \sum_{k=1}^K \mathbf{g}_{mk} x_k + \mathbf{n}_m \in \mathbb{C}^N, \quad (2)$$

with $x_k \sim \mathcal{N}_{\mathbb{C}}(0, p_k)$ being the uplink data symbol transmitted by UE k and $\mathbf{n}_m \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ representing the additive noise at AP m . Given a local combining vector $\mathbf{v}_{mk} \in \mathbb{C}^N$ for UE k , AP m forms a local soft estimate of x_k as

$$\hat{x}_{mk} = \mathbf{v}_{mk}^H \mathbf{y}_m = \mathbf{v}_{mk}^H \mathbf{g}_{mk} x_k + \sum_{l \neq k} \mathbf{v}_{mk}^H \mathbf{g}_{ml} x_l + \mathbf{v}_{mk}^H \mathbf{n}_m. \quad (3)$$

Then, based on the two-layer LSFD scheme [12, Sec. III-B], the APs forward $\{\hat{x}_{mk}, \forall m\}$ to the CPU, which then computes the final estimate via a second-layer linear combination $\check{x}_k = \sum_{m=1}^M a_{mk}^* \hat{x}_{mk}$, where a_{mk} is the complex scalar weight associated with AP m for UE k . The statistics-based LSFD weight vector for each UE can be obtained in closed-form from a Rayleigh quotient maximization, which both maximizes the use-and-then-forget (UatF) achievable SE and equivalently minimizes the conditional MSE, please see [20, Sec. II-A] for details.

In this paper, we move beyond the LSFD paradigm by allowing each AP to apply a matrix-valued weight for each UE, instead of using only a scalar second-layer coefficient. Specifically, we parameterize the weighted local combining vector for UE k at AP m as

$$\mathbf{v}_{mk} = \mathbf{W}_{mk} \check{\mathbf{v}}_{mk}, \quad (4)$$

where $\mathbf{W}_{mk} \in \mathbb{C}^{N \times N}$ is a statistics-based weighting matrix assigned to the AP-UE pair (m, k) , and $\check{\mathbf{v}}_{mk} \in \mathbb{C}^N$ is the instantaneous channel state information (CSI)-based local combining vector. Two commonly used choices for $\check{\mathbf{v}}_{mk}$ are the local MMSE (L-MMSE) combining $\check{\mathbf{v}}_{mk} = p_k (\sum_{l=1}^K p_l (\hat{\mathbf{g}}_{ml} \hat{\mathbf{g}}_{ml}^H + \mathbf{C}_{ml}) + \sigma^2 \mathbf{I}_N)^{-1} \hat{\mathbf{g}}_{mk}$ with $\mathbf{C}_{ml} = \mathbf{R}_{ml} - p_l \tau_p \mathbf{R}_{ml} \Psi_{ml}^{-1} \mathbf{R}_{ml}$ being the covariance matrix of the channel estimation error $\check{\mathbf{g}}_{ml} = \mathbf{g}_{ml} - \hat{\mathbf{g}}_{ml}$ and the local-maximum ratio (L-MR) combining $\check{\mathbf{v}}_{mk} = \hat{\mathbf{g}}_{mk}$. After local processing, the APs forward their soft estimates to the CPU, which forms the final detection statistic for UE k as

$$\check{x}_k = \sum_{m=1}^M \hat{x}_{mk} = \left(\sum_{m=1}^M \mathbf{v}_{mk}^H \mathbf{g}_{mk} \right) x_k + \sum_{l \neq k} \sum_{m=1}^M \mathbf{v}_{mk}^H \mathbf{g}_{ml} x_l + \mathbf{n}_k'. \quad (5)$$

Based on (5), the achievable SE of UE k is given by $\text{SE}_k = \frac{\tau_u}{\tau_c} \log_2(1 + \text{SINR}_k)$, where SINR_k is computed from the UatF bound as shown in (5) at the top of the page. Notably, under the structure in (4), an optimal set of weighting matrices

$$\text{SINR}_k = \frac{p_k \left| \sum_{m=1}^M \mathbb{E}\{\mathbf{v}_{mk}^H \mathbf{g}_{mk}\} \right|^2}{\sum_{l=1}^K p_l \mathbb{E}\left\{ \left| \sum_{m=1}^M \mathbf{v}_{mk}^H \mathbf{g}_{ml} \right|^2 \right\} - p_k \left| \sum_{m=1}^M \mathbb{E}\{\mathbf{v}_{mk}^H \mathbf{g}_{mk}\} \right|^2 + \sigma^2 \sum_{m=1}^M \mathbb{E}\{\|\mathbf{v}_{mk}\|^2\}} \quad (5)$$

$\{\mathbf{W}_{mk} : \forall m\}$ can be derived to maximize the achievable SE of UE k , as follows.

Proposition 1. *The optimal weighting matrices $\{\mathbf{W}_{mk} : \forall m\}$ that maximize the achievable SE can be obtained by first constructing a vectorized solution and then reshaping it. Specifically, we set $\mathbf{W}_{mk} = \text{vec}^{-1}(\mathbf{w}_{mk})$, where $\mathbf{w}_{mk} = [\mathbf{w}_k]_{(m-1)N^2+1:mN^2} \in \mathbb{C}^{N^2}$ denotes the sub-vector associated with AP m . By stacking all AP-specific sub-vectors, the collective vector for UE k is defined as $\mathbf{w}_k = [\mathbf{w}_{1k}^T, \mathbf{w}_{2k}^T, \dots, \mathbf{w}_{Mk}^T]^T \in \mathbb{C}^{MN^2}$. Then, the optimal statistics-based weighting vector for UE k is given by*

$$\mathbf{w}_k^* = \left(\sum_{l=1}^K p_l \mathbb{E}\{\mathbf{p}_{kl} \mathbf{p}_{kl}^H\} + \sigma^2 \mathbf{\Theta}_k \right)^{-1} \mathbb{E}\{\mathbf{q}_k\}, \quad (6)$$

and the corresponding maximized SINR is $\text{SINR}_k^* = \frac{p_k \mathbb{E}\{\mathbf{q}_k\}^H \left(\sum_{l=1}^K p_l \mathbb{E}\{\mathbf{p}_{kl} \mathbf{p}_{kl}^H\} + \sigma^2 \mathbf{\Theta}_k \right)^{-1} \mathbb{E}\{\mathbf{q}_k\}}{p_k \mathbb{E}\{\mathbf{q}_k\} \mathbb{E}\{\mathbf{q}_k\}^H + \sigma^2 \mathbf{\Theta}_k^{-1} \mathbb{E}\{\mathbf{q}_k\}}$, where $\mathbf{p}_{kl} = [\text{vec}(\mathbf{g}_{1l} \tilde{\mathbf{v}}_{1k}^H)^T, \dots, \text{vec}(\mathbf{g}_{Ml} \tilde{\mathbf{v}}_{Mk}^H)^T]^T \in \mathbb{C}^{MN^2}$, $\mathbf{\Theta}_k = \text{diag}\{(\mathbb{E}\{\tilde{\mathbf{v}}_{1k} \tilde{\mathbf{v}}_{1k}^H\} \otimes \mathbf{I}_N), \dots, (\mathbb{E}\{\tilde{\mathbf{v}}_{Mk} \tilde{\mathbf{v}}_{Mk}^H\} \otimes \mathbf{I}_N)\} \in \mathbb{C}^{MN^2 \times MN^2}$, and $\mathbf{q}_k = [\text{vec}(\mathbf{g}_{1k} \tilde{\mathbf{v}}_{1k}^H)^T, \dots, \text{vec}(\mathbf{g}_{Mk} \tilde{\mathbf{v}}_{Mk}^H)^T]^T \in \mathbb{C}^{MN^2}$.

Proof: We have $\sum_{m=1}^M \mathbb{E}\{\mathbf{v}_{mk}^H \mathbf{g}_{mk}\} = \sum_{m=1}^M \mathbb{E}\{\tilde{\mathbf{v}}_{mk}^H \mathbf{W}_{mk} \mathbf{g}_{mk}\} = \sum_{m=1}^M \mathbf{w}_{mk}^H \text{vec}(\mathbb{E}\{\mathbf{g}_{mk} \tilde{\mathbf{v}}_{mk}^H\})$ with step (a) based on [16, Eq. (32)] and [16, Eq. (34)]. Then, we can formulate $\mathbb{E}\{|\sum_{m=1}^M \mathbf{v}_{mk}^H \mathbf{g}_{mk}|^2\} = \mathbb{E}\{|\sum_{m=1}^M \text{tr}(\mathbf{W}_{mk}^H \mathbf{g}_{mk} \tilde{\mathbf{v}}_{mk}^H)|^2\} = \mathbb{E}\{|\sum_{m=1}^M \mathbf{w}_{mk}^H \text{vec}(\mathbf{g}_{mk} \tilde{\mathbf{v}}_{mk}^H)|^2\} = \mathbf{w}_k^H \mathbb{E}\{\mathbf{p}_{kl} \mathbf{p}_{kl}^H\} \mathbf{w}_k$, where step (a) and step (b) follow from [16, Eq. (32)] and [16, Eq. (34)], respectively. Finally, based on [16, Eq. (36)], we have $\sum_{m=1}^M \mathbb{E}\{\|\mathbf{v}_{mk}\|^2\} = \sum_{m=1}^M \text{tr}(\mathbf{W}_{mk}^H \mathbf{W}_{mk} \mathbb{E}\{\tilde{\mathbf{v}}_{mk} \tilde{\mathbf{v}}_{mk}^H\}) = \sum_{m=1}^M \mathbf{w}_{mk}^H (\mathbb{E}\{\tilde{\mathbf{v}}_{mk} \tilde{\mathbf{v}}_{mk}^H\}^T \otimes \mathbf{I}_N) \mathbf{w}_{mk} = \mathbf{w}_k^H \mathbf{\Theta}_k \mathbf{w}_k$. Thus, we can accordingly formulate SINR_k as

$$\text{SINR}_k = \frac{p_k |\mathbf{w}_k^H \mathbb{E}\{\mathbf{g}_k\}|^2}{\mathbf{w}_k^H \left(\sum_{l=1}^K p_l \mathbb{E}\{\mathbf{z}_{kl} \mathbf{z}_{kl}^H\} - p_k \mathbb{E}\{\mathbf{g}_k\} \mathbb{E}\{\mathbf{g}_k\}^H + \sigma^2 \mathbf{\Theta}_k \right) \mathbf{w}_k}. \quad (7)$$

Based on the standard Rayleigh quotient maximization theory [19, Lemma B.10] and steps in [19, C.3.2], we obtain the optimal $\mathbf{w}_k$ in (6) and the respective maximized SINR. ■

The OBE weighting scheme in (6) also minimizes the conditional MSE of the final decoding symbol for UE k as $\text{MSE}_k = \mathbb{E}\{|x_k - \tilde{x}_k|^2\}$. To prove this argument, by applying the weighted local combining structure in (4), the final decoding data in (5) can be compactly rewritten as $\tilde{x}_k =$

$\mathbf{w}_k^H \mathbf{q}_k x_k + \sum_{l \neq k} \mathbf{w}_k^H \mathbf{p}_{kl} x_l + \sum_{m=1}^M \tilde{\mathbf{v}}_{mk}^H \mathbf{W}_{mk} \mathbf{n}_m$, where the involved quantities are defined in Proposition 1. Then, we can expand $\text{MSE}_k = \mathbb{E}\{|x_k - \tilde{x}_k|^2\} = \mathbb{E}\{(x_k - \tilde{x}_k)(x_k - \tilde{x}_k)^H\} = \mathbb{E}\{x_k x_k^H\} - \mathbb{E}\{x_k \tilde{x}_k^H\} - \mathbb{E}\{\tilde{x}_k x_k^H\} + \mathbb{E}\{\tilde{x}_k \tilde{x}_k^H\}$. Using the second-order moments of x_k and the independence across UEs and conditioning on the channel statistics, we obtain

$$\mathbb{E}\{|x_k - \tilde{x}_k|^2\} = p_k - p_k \mathbb{E}\{\mathbf{q}_k^H\} \mathbf{w}_k - p_k \mathbf{w}_k^H \mathbb{E}\{\mathbf{q}_k\} + \mathbf{w}_k^H \left(\sum_{l=1}^K p_l \mathbb{E}\{\mathbf{p}_{kl} \mathbf{p}_{kl}^H\} \right) \mathbf{w}_k + \sigma^2 \mathbf{w}_k^H \mathbf{\Theta}_k \mathbf{w}_k. \quad (8)$$

By taking the partial derivative of MSE_k with respect to $\mathbf{w}_k^*$ and setting it to zero, the MSE-minimizing solution follows as $\mathbf{w}_k = p_k \left(\sum_{l=1}^K p_l \mathbb{E}\{\mathbf{z}_{kl} \mathbf{z}_{kl}^H\} + \sigma^2 \mathbf{\Theta}_k \right)^{-1} \mathbb{E}\{\mathbf{g}_k\}$, which coincides with the OBE weighting solution in (6). Thus, the OBE weighting scheme in (6) can not only maximize the UatF-bound evaluated achievable SE for UE k , but also minimize the conditional MSE of the final detected symbol for UE k .

IV. NUMERICAL RESULTS

In this section, we evaluate the proposed scheme in a CF mMIMO setup with a $200 \times 200 \text{ m}^2$ coverage area, where the center of the service region is placed at the origin. The APs are mounted on a wall located on the x - y plane with the z -coordinate fixed to -100 . Half-wavelength antenna spacing is adopted, i.e., $\Delta_H = \Delta_V = \bar{\lambda}/2$, where $\bar{\lambda}$ denotes the carrier wavelength and the carrier frequency is $f_c = 1.9 \text{ GHz}$. For the channel model, the LoS component $\bar{\mathbf{g}}_{mk}$ is generated using a spherical-wave formulation [21], [22] as

$$\bar{\mathbf{g}}_{mk} = \sqrt{\beta_{mk}^{\text{LoS}}} [e^{-j \frac{2\pi}{\bar{\lambda}} (d_{m1k} - d_{mk})}, \dots, e^{-j \frac{2\pi}{\bar{\lambda}} (d_{mNk} - d_{mk})}]^T, \quad (9)$$

where $d_{mnk} = \|\mathbf{r}_{mn} - \mathbf{s}_k\|$ is the distance between the n -th antenna of AP m and UE k and $d_{mk} = \|\mathbf{r}_m - \mathbf{s}_k\|$ is the reference distance. Moreover, β_{mk}^{LoS} denotes the large-scale fading coefficient associated with the LoS component. For the NLoS component, an isotropic scattering environment is assumed and the spatial correlation matrix $\mathbf{R}_{mk}$ is constructed according to [23], where its (n_1, n_2) -th entry is given by $[\mathbf{R}_{mk}]_{n_1 n_2} = \beta_{mk}^{\text{NLoS}} \text{sinc}(2 \|\mathbf{r}_{mn_1} - \mathbf{r}_{mn_2}\| / \bar{\lambda})$ with $\mathbf{r}_{mn_1}$ and $\mathbf{r}_{mn_2}$ being the coordinates of the n_1 -th and n_2 -th antennas of the m -th array, respectively, and β_{mk}^{NLoS} being the large-scale fading coefficient of the NLoS component. Besides, we can compute $\beta_{mk}^{\text{LoS}} = \frac{\kappa_{mk}}{\kappa_{mk} + 1} \beta_{mk}$, $\beta_{mk}^{\text{NLoS}} = \frac{1}{\kappa_{mk} + 1} \beta_{mk}$, where the Rician factor is modeled as $\kappa_{mk} = 10^{1.3 - 0.003 d_{mk}}$. The overall large-scale fading coefficient β_{mk} is generated using the 3GPP COST 231 Walfish-Ikegami model in [24, Eq. (5.2-3)]. In addition, unless otherwise specified, we set $L_{AP} =$

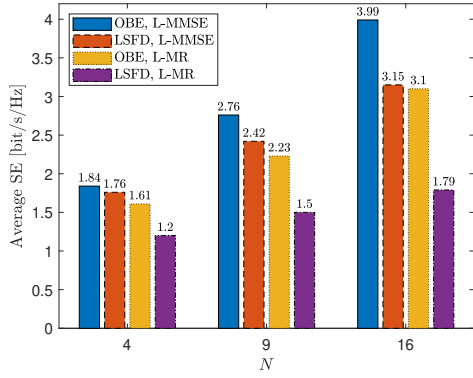


Fig. 1. Average SE per UE versus N for the proposed OBE and conventional LSFD weighting schemes with $M = 10$ and $K = 10$.

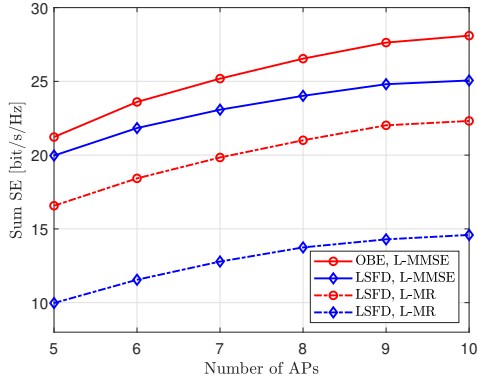


Fig. 2. Average SE per UE versus M for the OBE weighting and LSFD weighting schemes with $K = 10$ and $N = 9$.

12.5 m, $L_{UE} = 1.5$ m, the bandwidth as 20 MHz, transmit power $p_k = 0.2$ W, the coherence block length $\tau_c = 200$, pilot length $\tau_p = 5$, and the noise power $\sigma^2 = -94$ dBm.

Fig. 1 compares the average per-UE SE achieved by the proposed OBE weighting and the conventional LSFD weighting for different values of N . The OBE scheme consistently delivers higher SEs than LSFD, and the performance gap widens as N increases. This trend stems from the fact that OBE enables antenna-element-aware weighting and can better capitalize on the additional spatial processing capability offered by larger arrays, whereas LSFD applies only a single scalar weight to all antennas within an AP. In addition, the SE gain brought by OBE is more pronounced when combined with L-MR than with L-MMSE. Since L-MR provides no interference suppression, OBE can effectively compensate for this weakness through enhanced system-level weighting, leading to a larger relative improvement.

Fig. 2 shows the average SE per UE versus M . We observe that, as N increases, all schemes achieve higher SE, but with diminishing marginal gains, reflecting the saturation of cooperation benefits. The proposed OBE consistently outperforms LSFD across the entire considered range of M , showing that matrix-valued weighting can better exploit the available spatial and cooperation processing capability. The improvement achieved by OBE is also particularly pronounced under L-MR. Remarkably, L-MR combined with OBE attains

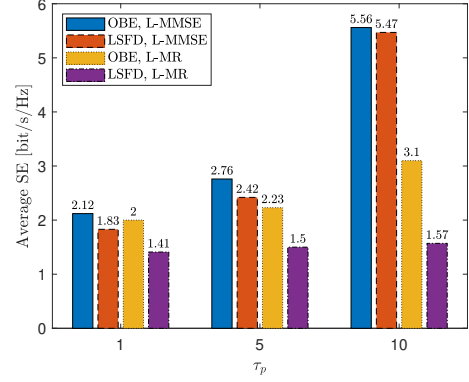


Fig. 3. Average per-UE SE versus τ_p for the OBE and LSFD weighting schemes over $M = 10$, $K = 10$, and $N = 9$.

SE close to that of the L-MMSE with LSFD, highlighting that strong system-level weighting can substantially narrow the performance gap between low- and high-complexity local processing combining schemes.

Fig. 3 examines how the pilot length τ_p affects the achievable SE. Increasing τ_p can mitigate pilot contamination and improve the channel estimation quality. For L-MMSE combining, the SE difference between OBE and LSFD gradually narrows as τ_p grows, since L-MMSE already provides strong interference suppression and benefits substantially from more accurate CSI, leaving less room for additional gains from the weighting schemes. This also suggests that OBE is most advantageous when pilot contamination is more severe. In contrast, under L-MR combining, the performance gap between OBE and LSFD expands with larger τ_p , indicating that OBE can exploit the improved CSI more effectively than LSFD when the local combining scheme has limited interference suppression capability. Nevertheless, even with OBE, L-MR does not catch up with L-MMSE at large τ_p , where the performance gap between L-MR and L-MMSE increases as CSI becomes more accurate. This reveals that once estimation errors are sufficiently reduced, the ultimate performance is largely determined by the interference suppression capability of the local combining scheme.

V. CONCLUSIONS

In this paper, we studied an antenna-aware matrix-valued OBE weighting strategy for CF mMIMO networks by assigning a long-term weight matrix to each AP-UE pair. Compared to the conventional LSFD, the OBE weighting scheme can exploit the intra-AP processing capability of multi-antenna arrays to enhance the SE. The SE improvement achieved by the OBE scheme compared to the LSFD scheme increased with the number of antennas, as larger arrays provided a richer spatial structure that could be efficiently leveraged only with antenna-aware weighting. Besides, the improvement is more pronounced for the L-MR combining than for the L-MMSE combining, indicating that antenna-aware weighting is particularly effective in compensating for the limited interference suppression of the low-complexity local combining scheme.

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